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What Determines the Sucess and the Failure of Evolutionary Strategies (ES) in LLM fine-tuning?

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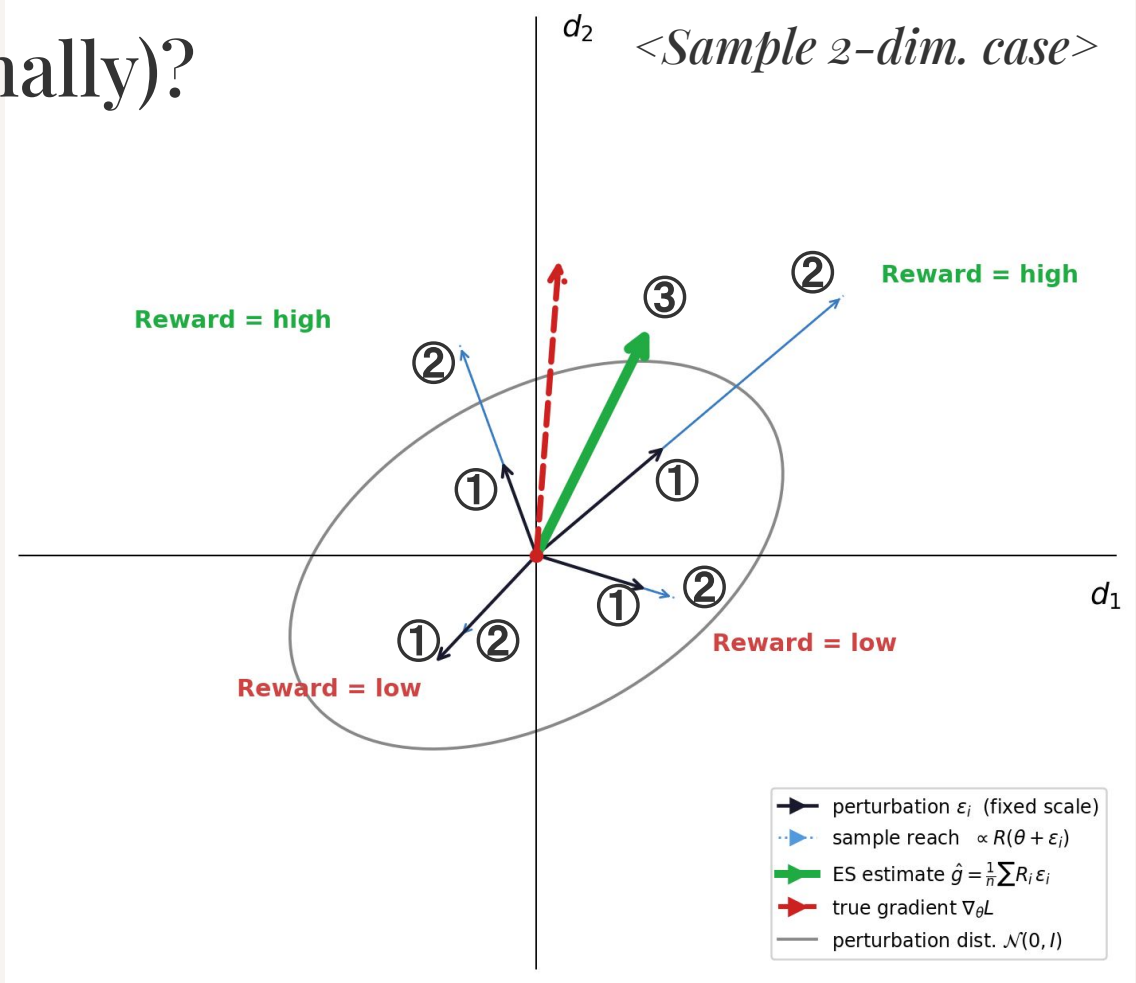
What is ES (Informally)?

- A zeroth-order optimization algorithm that works completely different from first-orders like GD, SGD, Adam, PPO, etc.
- ① Throw random variations of your policy into the world, see which ones do better
- ② Give more credit to the directions that scored high rewards and less to the ones that flopped
- ③ Average those weighted directions → you get a "sense" of which way to improve
- Repeat. That's it. No gradients, no backprop, just reward signals



What is ES (Formally)?

$$\hat{g} = \frac{1}{N\sigma} \sum_{i=1}^{N} \left[R(\theta + \sigma\varepsilon_i) - R(\theta - \sigma\varepsilon_i) \right] \varepsilon_i$$

$$\theta_{t+1} = \theta_t - \eta \hat{g}$$

$\hat{g}$: gradient estimate

N : population (number of random perturbations)

$R(\cdot)$: reward function

θ : previous LLM weights

σ : perturbation scale (e.g., $\sigma = 0.001$)

$\varepsilon_i \sim \mathcal{N}(0, I)$: random perturbation

A brief background of ES

1950s/60s - 2000s: Origination

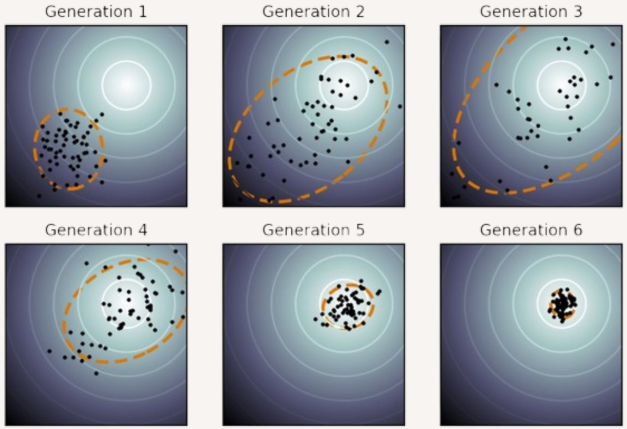
- ES was originated from genetic algorithms in 1960s-70s as a black-box optimization
- CMA-ES (2001) and Natural ES (2014)

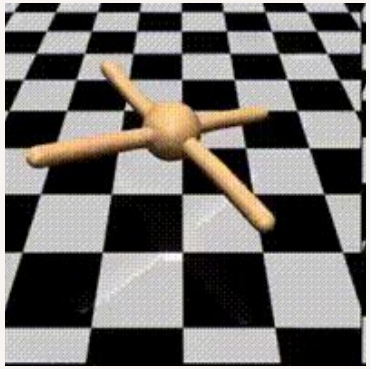
2017: OpenAI made ES hot again

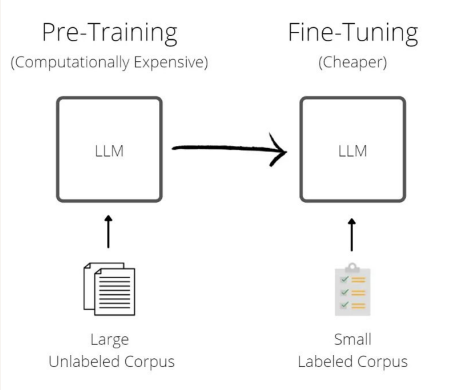
- ES becomes "modern" in ML by scaling massively on RL benchmarks (Atari/MuJoCo)
- Showed zeroth-order methods like ES can match complex RL techniques

2023-Present : LLM fine-tuning applications

- Active discussions about ES in LLM fine-tuning
- ES was recognized for its viability in (1) black-box models (2) memory-efficiency (3) non-differentiable /discrete rewards







Source: Hansen, N., & Ostermeier, A. (2001). Completely derandomized self-adaptation in evolution strategies. Evolutionary Computation, 9(2), 159–195; Wierstra, D., et al. (2014). Natural evolution strategies. Journal of Machine Learning Research, 15(1), 949–980.; Salimans, T., et al. (2017). Evolution strategies as a scalable alternative to reinforcement learning. *arXiv:1703.03864;* Qiu et al. (2025). Evolution strategies at scale: LLM fine-tuning beyond reinforcement learning. *arXiv:2509.24372.*

Lesson 1 Backdrop: Which N (population) to use?

Larger N gives a more accurate gradient estimate, but at greater compute cost

Both papers below used the same ES algorithm yet reached opposite conclusions about N — why?

MeZO (Malladi et al., 2023)

ng training, n can be treated as a hyperparameter and
xperiments (Appendix A), $n = 1$ is the most efficient.

ES at Scale (Qiu et al., 2025)

is that the ES implementation only needs a population of 30 to effectively optimize billions of parameters. In con-

Source: Malladi et al. (2023). Fine-Tuning Language Models with Just Forward Passes. NeurIPS 2023 (oral). *arXiv:2305.17333*; Qiu et al. (2025). Evolution strategies at scale: LLM fine-tuning beyond reinforcement learning. *arXiv:2509.24372*.

Lesson 1: Different Reward Requires Different Population

We attribute both papers' paradoxical conclusions about N to their use of fundamentally different reward functions

Cross-Entropy (CE) Reward Function

$$R(x, y^*) = \log P_\theta(y^* \mid x)$$

- Continuous
- Dense gradient signal (magnitude represents confidence)
- Low variance and high sample efficiency
- **Requires white box logit access**

Binary Reward Function

$$R(y, y^*) = \begin{cases} 1 & \text{if } y = y^* \\ 0 & \text{otherwise} \end{cases}$$

- Discrete
- Sparse gradient signal (only direction, no magnitude)
- High variance and low sample efficiency
- **Generation only - black box**

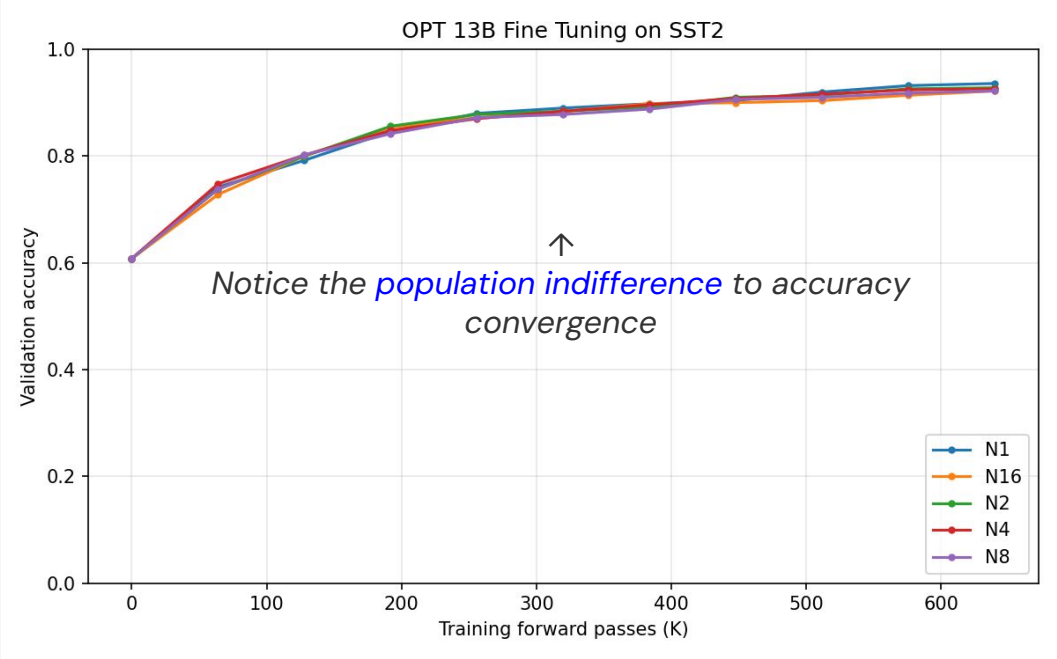
CE reward works for all N… but is it actually realistic?

Cross-Entropy Loss Function

- Works for all N. CE is essentially N-agnostic
- CE reward is smooth, so every perturbation produces a nonzero gradient signal, making per-forward-pass efficiency N-independent.
- Very favorable performance

Limitation

- CE is essentially paradoxical with the biggest advantage of ES: black-box capability
- Black-box assumption is broken since gaining logit assumes white-box access



** What is SST2?: Binary movie review sentiment classification (positive/negative)*
e.g. Q: "The film was a delightful, heartwarming adventure." → A: positive

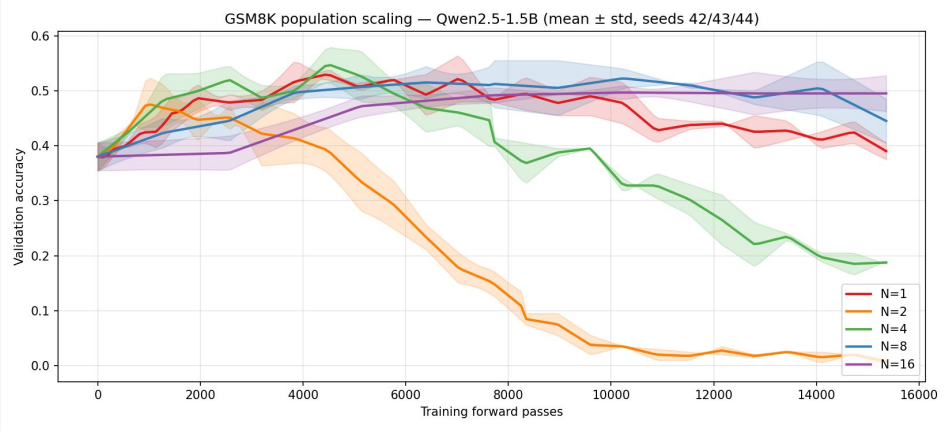
Binary reward is more realistic but does NOT work for all N

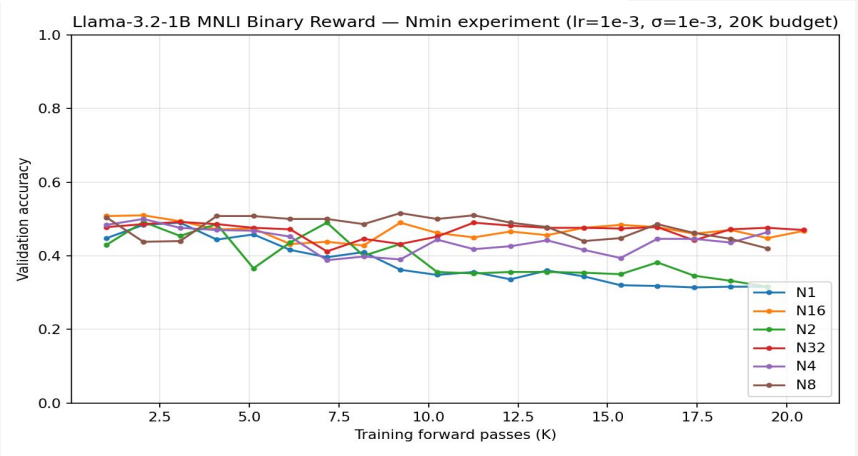
Binary Reward Function

- Not N agnostic - we observe that the model is likely to degrade at low N
- This is due to how binary rewards don't provide a clear signal on how the weights should be perturbed

The Pros/Cons of Binary Rewards

- Binary rewards are true to the original intent of evolutionary strategies
- BUT ES with binary reward doesn't always work





** What is GSM8K?: Grade-school math word problems requiring multi-step arithmetic reasoning.*
e.g. Q: "Janet has 3 apples and buys 5 more. She gives 2 away. How many does she have?" → A: 6

** What is MNLI?: Natural language inference: does a premise entail, contradict, or stay neutral to a hypothesis?*
E.g. Premise: "The cat sat on the mat." Hypothesis: "An animal is on the mat." → A: entailment

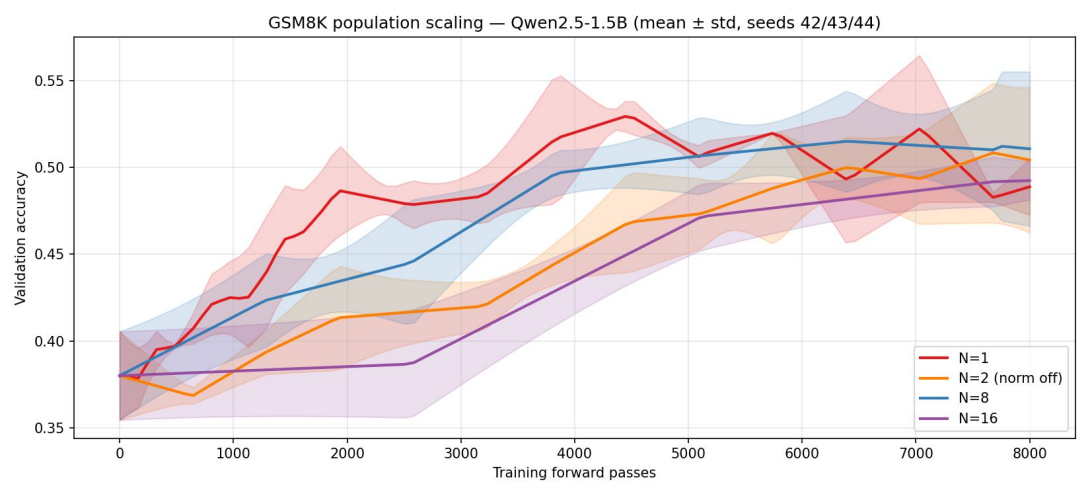
Lesson 2: Vanilla Model's Failure Mode Predicts Performance

The type of error a pre-trained model makes — not just a raw performance score — determines whether ES fine-tuning can succeed.

	ES-Benign	ES-Malignant
Error type	Incorrect but task-aware	Fundamental incapacity
What it looks like	Shows reasoning steps, arrives at the wrong number	Says "no" to every question, or just repeats the input prompt
Why errors arise	Model understands the format but lacks knowledge or makes reasoning mistakes	Model cannot parse or engage with the task at all
Correctable by ES?	Yes — errors share a direction across examples	No — errors are structurally irreducible

ES-Benign Regime: Vanilla is Capable

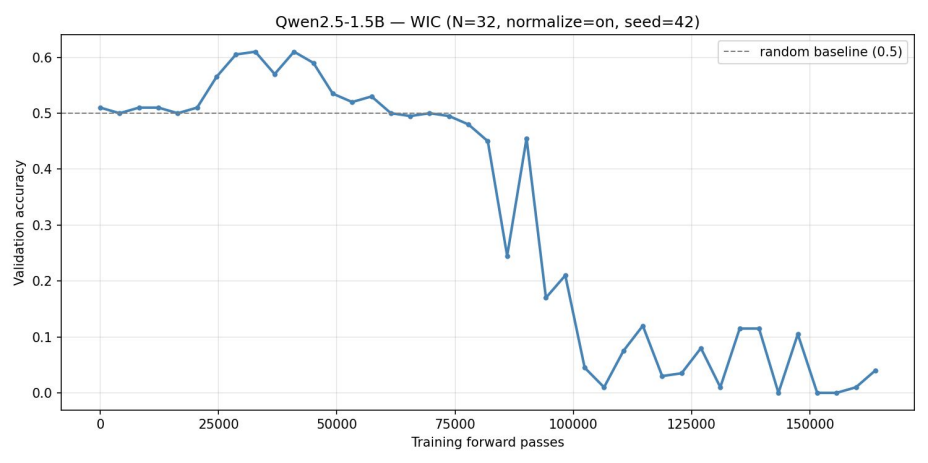
- When a model has the right format but wrong answer, small perturbations produce meaningfully different rewards — giving ES a real gradient signal.
- One perturbation direction consistently scores higher → update points somewhere useful
- The reward landscape has genuine slope: model understands the task, just needs to be nudged



	✅ Correct	❌ Wrong
Question	A store sells 3 pens for $4. How much do 12 pens cost?	
Model output	*1 pen = $4/3. 12 pens = 12 × $4/3 = $16.* ***#### 16***	*3 pens = $4, so 12 pens = 4 × $4 = $12.* ***#### 12***
Gold	#### 16	
Failure type	—	Wrong arithmetic, correct format

ES-Malignant Regime: Vanilla is Dumb

- When a model is fundamentally incapable, per-example errors are inconsistent
  - they cancel at the batch level, leaving ES with no useful gradient.
- No gradient → each update is a random walk with no consistent progress
- Random drift self-reinforces
  - a slight bias toward "no" gets rewarded on "no" examples → model completely collapses with parsing errors



	Predicted: No	Predicted: Yes
Gold: No (49.5% of data)	96.0%	4.0%
Gold: Yes (50.5% of data)	93.1%	6.9%

** What is WIC?: Word-in-Context: does a target word carry the same meaning in two different sentences?*
e.g. S1: "I need to book a flight." S2: "She read the book." → A: False (different senses)

Suggesting the way to decide the regime for practictioners

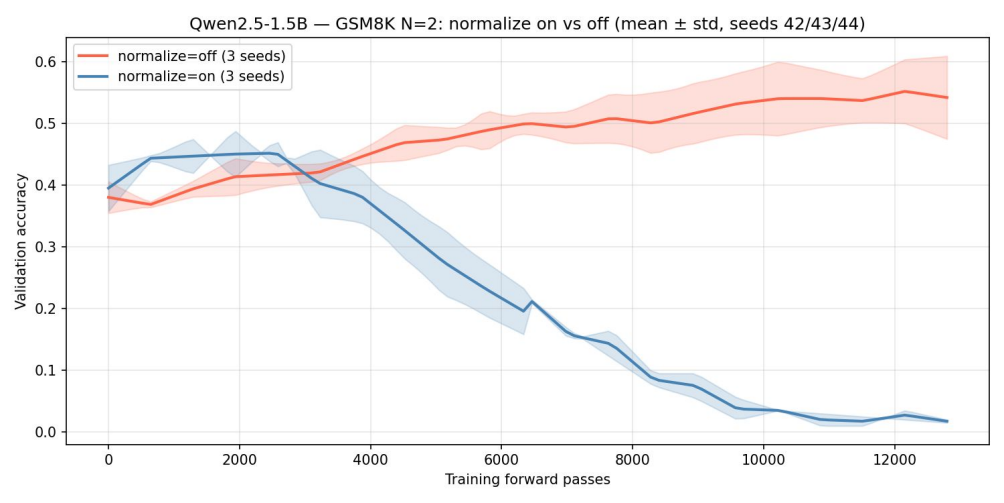
- You can identify the regime by inspecting the vanilla model's failure mode directly
- For a more systematic, numerical signal: we introduce ρ (rho) — the correlation between rewards under two antithetic perturbations of the same model
- ES-benign: ρ lands in a moderate range (0.3–0.7) — ES-malignant: ρ pushes to extremes (very low = chaotic; very high = frozen)

$$\rho_{\text{corr}} = \text{Corr}(R(\theta + \sigma\varepsilon_j),\ R(\theta - \sigma\varepsilon_j))$$

	LLaMA-1B / MNLI	Qwen-1.5B / GSM8K	Qwen-1.5B / WIC
ρ value	0.14	0.42	0.88
ρ regime	Very low	Moderate	Very high
Failure mode	Chaotic — repeats prompt, random outputs	Capable but imperfect — wrong numbers, right format	Frozen — says "no" to everything
ES outcome	Malignant	Benign ✓	Malignant

Lesson 3: Reward Normalization Sometimes Hurts!

- Advantage normalization is a standard practice in ES and GRPO (default in most papers)
- We've kept observing catastrophic failure in N=2 and attributed it to normalization
- At N=2, z-score normalization amplifies noise into unit-magnitude random updates
  - Signal is weak relative to batch noise → denominator is noise-dominated → scale information destroyed
- Norm OFF restores the signal filter: advantage magnitude is preserved
- No prior paper has ablated this
  - may explain why small N failed



Normalization ON	Normalization OFF
A = +0.06 → ±0.71 (noise magnitude)	A = +0.06 → stays → +0.06
A = +0.50 → ±0.71 (same as weak signal!)	A = +0.50 → stays → +0.50
Scale information lost	**Scale information preserved**

Conclusion, Limitations & Next Steps

- **Three Takeaways**
  - 1. Different Reward Requires Different Population
  - 2. Vanilla Model's Failure Mode Predicts Performance
  - 3. Normalization Sometimes Hurts
- **Limitations**
  - broader coverage needed to validate the benign/malignant taxonomy
  - Single seed for normalization ablations; multi-seed averaging only on population scaling
- **Next Steps**
  - Would Small N work in binary rewards?
    - Since we discovered normalization might hurt, is there way we can make N<30 work?
  - Creating a new reward function
    - More realistic than CE but also more informative than Binary
    - e.g., partial credit, soft accuracy
    - a middle ground that may lower N_min without requiring gradient access and thus achieve higher computational efficiency

THANK YOU

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